Comparing Yeast Metabolic Rates

Which sugar does yeast "eat" fastest? A balloon experiment

AIM
To compare how fast yeast uses different sugars (and one artificial sweetener) by measuring the carbon dioxide gas it releases.

Background

Yeast is a single-celled fungus. When it has no oxygen to spare, it gets energy from sugar by fermentation, releasing carbon dioxide and alcohol:

glucose $\rightarrow$ 2 ethanol + 2 carbon dioxide + energy
$C_6H_{12}O_6 \rightarrow 2\ C_2H_5OH + 2\ CO_2$

The carbon dioxide gas makes bubbles in the liquid and inflates a balloon fixed over the bottle. The faster the balloon inflates, the faster the yeast is working (the faster its metabolic rate). The sugars differ: glucose and fructose are simple sugars the yeast can use directly; sucrose must first be split into glucose and fructose by a yeast enzyme (invertase); aspartame is an artificial sweetener that yeast cannot ferment for energy.

Hypothesis

Before you start, write a prediction: "I think the balloon on the _____ bottle will inflate fastest, because _____."

Materials

- 5 identical clean bottles (500 mL plastic bottles work well)
- 5 balloons of the same size and type
- glucose, fructose, sucrose (table sugar), aspartame (powder), baker's dried yeast
- lukewarm water (about 35–40 °C), measuring jug, spoons, a balance if available
- marker, tape, a string and a ruler (to measure the balloon), a timer

MAKE IT A FAIR TEST

- Use the **same amount** of each sugar. Spoons hold different masses of different powders, so weigh them if possible (e.g. 10 g each).
- Use the **same volume** of water at the same temperature, and the same amount of yeast in every bottle.
- Add a **fifth bottle with yeast and water only** (no sugar) as the **control**. It shows what yeast does when it has nothing to eat.
- Do not change anything except the sugar.

Procedure

1. Label the bottles: **Glucose, Fructose, Sucrose, Aspartame, Control (no sugar)**.
2. Pour the same amount (e.g. 200 mL) of lukewarm water into each bottle. *Hot water above about 45 °C will kill the yeast*.
3. Add the sugar to each bottle (none to the control) and swirl to dissolve.
4. Add the same amount of yeast (e.g. 1 tablespoon, or 7 g) to each bottle. Swirl gently.
5. Immediately stretch a balloon over the mouth of each bottle. Make sure it is tight and **starts empty**.
6. Start the timer. Every **10 minutes for 60 minutes**, measure around the widest part of each balloon with the string, then measure the string with a ruler. Record the circumference in centimetres.
7. Also record what you see: bubbles, foam, smell.

Results

Time (min)	Glucose	Fructose	Sucrose	Aspartame	Control
0	0	0	0	0	0
10					
20					
30					
40					
50					
60					

Balloon circumference (cm).

Graph: plot time on the x-axis and circumference on the y-axis, one line per bottle. The steepest line shows the fastest metabolic rate.

Discussion

1. Which medium gave the fastest inflation? The slowest? Was your hypothesis correct?
2. What happened in the control bottle? Why is a control important?
3. The aspartame bottle: what does the result tell you about aspartame and yeast?
4. Sucrose is made of glucose and fructose joined together. Why might it start more slowly than glucose?
5. What could have made the experiment unfair? (Different amounts, temperature, yeast freshness, balloon size.) How would you improve it?

NOTES FOR TEACHERS

Results vary with the yeast, temperature and bottle size, so do not tell students the "right answer" beforehand. Typical findings: glucose, fructose and sucrose all inflate their balloons (sucrose may lag at the start), while aspartame behaves like the no-sugar control. Extension ideas: vary the temperature (cold, warm, hot water), the amount of sugar, or test honey, jaggery and other sweeteners.

SAFETY

Do not eat or drink anything used in the experiment. Do not blow up the balloons by mouth. Wipe spills — sugar water is sticky and attracts insects. Dispose of the contents down the drain with plenty of water.